

Syllabus

Spring semester 2026

Course Title	Calculus 2		Course Code		MC3109		Credit	4	Classroom	형설관3407
			Type		elective		Hours	5		
Teacher	Name	Class no.	Office	Office hours		Contact		E-mail		
	Jung Eun Kim	EC2	3503	Wed 2,3, Fri 2		606-2386		Jkim.math@ksa.hs.kr		
	Jerome Tambour	EC1	3507	Mon 6,7 Fri 3		606-2148		jerome@ksa.hs.kr		

1. Course Descriptions

Course Objectives	
◆ To introduce students to understand integration, differential equations, series, derivatives and integrals of vector functions and to solve real world problem related to them.	
Contents	
1. Further Applications of Integration i. Arc length ii. Area of a Surface of Revolution iii. Applications to Physics and Engineering iv. Probability 2. Differential Equations i. Modelling with Differential Equations ii. Direction Fields and Euler's Method iii. Separable Equations iv. Models for Population Growth v. Linear Equations 3. Parametric Equations and Polar Coordinates i. Curves Defined by Parametric Equations ii. Calculus with Parametric Curves iii. Polar Coordinates iv. Areas and Lengths in Polar Coordinates v. Conic Sections vi. Conic Sections in Polar Coordinates	4. Infinite Sequences and Series i. Sequences ii. Series iii. The Integral Test and Estimates of Sums iv. The Comparison Tests i. Alternating Series and Absolute Convergence v. The Ratio and Root Tests vi. Strategy for Testing Series vii. Power Series viii. Representations for Functions as Power Series ix. Taylor and Maclaurin Series x. Applications of Taylor Polynomials 5. Vector Functions i. Vector Functions and Space Curves ii. Derivatives and Integrals of Vector Functions iii. Arc Length and Curvature iv. Motion in Space: Velocity and Acceleration

2. Core Target Competencies

Subject Core Competencies						
Comprehension Adaptability	Problem- Solving Ability	Communication	Analytical Reasoning	Creative Thinking	Integrative Thinking	Application Ability
Comprehension, Adaptability, Problem-Solving Ability (Assignments, Performance Assessment): Understanding basic concepts of the subject and applying learned content appropriately in given situations. Communication, Analytical Reasoning, Creative Thinking (Assignments, Performance Assessment, Written Exams): Evaluating the ability to analyze problems, creatively solve them by applying learned content, and logically articulate the solutions. Integrative Thinking, Application Ability (Assignments, Written Exams): Assessing the ability to solve problems in various fields such as physics, statistics, and geometry by utilizing calculus.						

3. Textbooks and References

- A. Textbook: Calculus, James Stewart, D. Clegg, S. Watson, 9th Edition, ISBN 978-0-357-11346-2
 B. References: Thomas' CALCULUS, Finney, Weir and Giordano, ISBN 0134438981

4. Class Method

- A: Lecture
 B: Solving problems
 C: Presentation

5. Evaluation

Activities & Participation	Quizzes	Mid-term Exam	Final Exam	Attendance & Attitude	Total
8%	12%	35%	35%	10%	100%
40	60	175	175	50	500

1) Activities

Activities include homework, in-class assignments and presentations.

2) Quizzes

There will be two in-class quizzes.

3) Exams

Mid-term : Chapter 8~10 Final : Chapter 11, 13 (It can be changed.)

3) Attendance

Absence: -3 points except for illness or official cases

(with confirmation from our school infirmary and contact in advance)

Lateness : -1 point

6. Criteria for Awarding Credits

☞ **Students with excellent academic achievement are recommended for the Academic Excellence Award and the Principal's Award, and are recorded according to core competencies in the Subject Details. However, students with poor academic attitude or poor attendance are excluded.**

● **Attendance for at least three-quarters of the total class hours is required to earn credit.**

● **Calculation of the basis score for awarding credit to absentees:** The score for an exam missed due to an official or medical absence is determined by taking the student's percentile score from their other exams in this subject. We find the score which corresponds to the same percentile for the missed exam. This score is then multiplied by a factor based on the reason for absence (100% for an official absence, 80% for a medical absence).

● **If a basis score cannot be calculated, the grading method:** If there is no basis score available, a make-up examination shall be conducted.

7. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Course introduction 8.1 Arc length 8.2 Area of a Surface of Revolution	
Core Competencies	♦ Problem solving ability, application ability	
Week 2 3.2.~3.8.	8.3. Applications to Physics and Engineering 8.5. Probability (Random variables, Expectation)	3.2.(Mon) Holiday
Core Competencies	♦ Problem solving ability, application ability, integrative thinking	
Week 3 3.9.~3.15.	8.5 Probability (Variance, Bernoulli distributions, Normal distributions)	
Core Competencies	♦ Application ability, integrative thinking, analytical reasoning	
Week 4 3.16.~3.22.	9.1 Modelling with Differential Equations 9.2 Direction Fields and Euler's Method 9.3 Separable Equations	
Core Competencies	♦ Comprehension adaptability, communication	
Week 5 3.23.~3.29.	9.4 Models for Population Growth 9.5 Linear Equations 10.1 Curves Defined by Parametric Equations	Quiz 1 (Chapter 8)
Core Competencies	♦ Creative thinking, comprehension adaptability	
Week 6 3.30.~4.5.	10.2 Calculus with Parametric Curves 10.3 Polar Coordinates 10.4 Areas and Lengths in Polar Coordinates	
Core Competencies	♦ Analytical reasoning, problem solving ability	
Week 7 4.6.~4.12.	10.5 Conic Sections 10.6 Conic Sections in Polar Coordinates Review	
Core Competencies	♦ Comprehension adaptability, communication	

Week 8 4.13.~4.19.	Midterm Exams (4.14~4.17.)	4.13.(Mon) Regular Classes
Week 9 4.20.~4.26.	11.1 Sequences 11.2 Series	
Week 10 4.27.~5.3.	11.3 The Integral Test and Estimates of Sums 11.4 The Comparison Tests	
Core Competencies	◆ Comprehension adaptability, analytical reasoning	
Week 11 5.4.~5.10.	11.5 Alternating Series and Absolute Convergence 11.6 The Ratio and Root Tests	
Core Competencies	◆ Application ability, problem solving ability	
Week 12 5.11.~5.17.	11.7 Strategy for Testing Series 11.8 Power Series	
Core Competencies	◆ Application ability, problem solving ability	
Week 13 5.18.~5.24.	11.9 Representations for Functions as Power Series 11.10 Taylor and Maclaurin Series	Quiz 2 (11.1~8)
Core Competencies	◆ Creative thinking, integrative thinking	
Week 14 5.25.~5.31.	11.11 Applications of Taylor Polynomials 13.1 Vector Functions and Space Curves	5.25.(Mon) Holiday
Core Competencies	◆ Creative thinking, integrative thinking	
Week 15 6.1.~6.7.	13.2 Derivatives and Integrals of Vector Functions 13.3 Arc Length and Curvature 13.4 Motion in Space: Velocity and Acceleration	6.3.(Wed) Election day
Core Competencies	◆ Application ability, analytical reasoning	
Week 16 6.8.~6.14.	Review	Make-up class week
Week 17 6.15.~6.19.	Final Exams (6.18~6.19.)	

※ The above plans may change according to the academic schedule.

8. Core Problem List

Chapter	Problem Number
8.1	9, 13, 17, 39, 53
8.2	10, 19, 20, 33, 37, 39, 46
P613	Discovery project 1, 2, 3, 4, 5
8.3	(Example 1, 2) 26, 31, 44, 47, 48, 50
8.5	17, 18, 20
P638~639	4, 9, 12, 14, 21, 23
P640~641	1, 2, 4, 5, 9, 10
9.1	13, 15, 19, 21, 23, 29
9.2	2, 11, 18, 22, 28
9.3	7, 19, 24, 35, 41, 43, 57
9.4	5, 7, 9, 11, 13, 19, 25
9.5	8, 10, 11, 19, 23, 27, 28, 38
P694	Concept Check 5, 6, 7, True-False Quiz 2, 7, 9
P695~P696	4, 6, 10, 13, 18, 20
P697~698	3, 4, 9, 13
10.1	14, 19, 20, 21, 27, 28, 30, 34, 51, 52, 56, Lab. project 1
10.2	9, 12, 20, 23, 35, 42, 49, 56, 72, 77, 79, 80, 81, 84
10.3	3, 39, 43, 46, 47, 48,
10.4	4, 8, 13, 21, 22, 27, 29, 31, 32, 35, 36, 39, 40, 42, 43, 44, 54, 67, 70, 75, 76
10.5	58, 59, 63, 65, 66
10.6	17, 21, 22, 27, 30,
P760	1, 2, 4, 5, 6
11.1	74, 76, 85, 89, 90, 95, 97, 98
11.2	61, 66, 70, 72, 77, 83, 84, 85, 86, 87, 88, 89, 90, 93, 95, 96
11.3	32, 34, 46, 47
11.4	32, 33, 34, 37, 40, 45, 46, 47, 48, 49, 50, 51
11.5	29, 40, 46, 48, 49, 50
11.6	17, 24, 25, 35, 40, 41, 45, 46
11.7	2, 4, 6, ..., 48 (problems numbered even)
11.8	17, 19, 34, 35, 36, 37, 38, 39, 40, 41, 43, 45
11.9	1, 2, 14, 15, 38, 40, 41, 42, 46, 47, 50
11.10	2, 3, 4, 5, 11, 14, 16, 23, 30, 48, 61, 62, 79, 81, 83, 84, 96, 97, 98
11.11	5, 15, 19, 30, 39
P859-860	5, 15, 9, 30, 39
P861	59, 60(a), 62
P863-866	1, 2, 4, 5, 7, 8, 11, 12, 14, 15, 16, 20, 22, 23
13.1	1, 4, 9, 13, 15, 21~26. 35, 36, 51, 55, 58, 61(c)
13.2	1, 3, 8, 18, 23, 26, 41, 43, 51, 55
13.3	4, 6, 14, 16, 18, 19, 26, 34, 46, 54, 64, 73
13.4	1, 9, 17, 21, 25, 29, 30, 33, 39, 45
P965	True-False 6, 8, 9, 13
P966~967	9, 11, 19, 23, 24
P968~970	1, 2, 3, 4, 5, 7, 9